

The Storyline

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Full Research Guidelines are:

<https://jvgemert.github.io/ResearchGuidelinesInDL.pdf>

1 Research as a creative process

My love for doing scientific research is in its creative processes. Exploring uncharted territory, brainstorming, critical reflections, coming up with research questions and rephrasing them, creating precisely controlled experiments, how to keep everything logically consistent, puzzling over experimental results, choosing which results to keep and which results are not relevant, writing a structured narrative, creating high entropy figures, how to best present and concisely communicate the key insights of the work, etc. Creative processes, however, are acutely different from technical crafts, as eruditely put by music producer Rick Rubin [1]. In the field of machine/deep learning these crafts involve technical skills such as math, programming, software engineering, statistical methodologies, etc. These skills are not only essential to get a timely, rigorous, and trustworthy answer, they are also important to detail in a publication, so that they can be scrutinized, questioned, replicated, and built upon. Yet, doing research is not only applying skills to find an answer, it is often not even clear what the question is [2]. Thus, it is normal (essential?) that the research direction changes/evolves many times, see Fig 1; this is the creative process at work: doing the work to find the question. This document is about catalyzing the scientific creative process in search of the question.

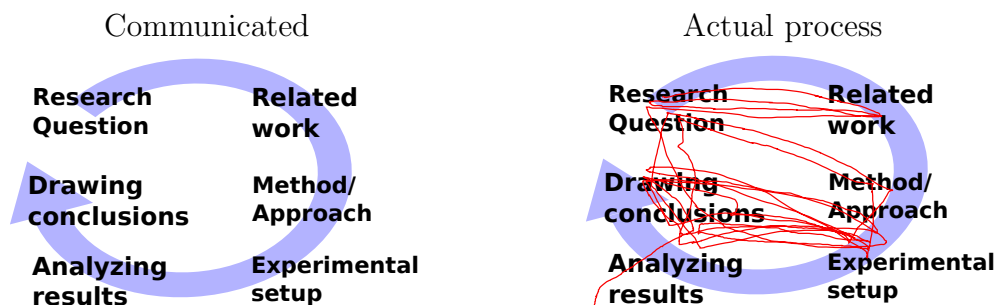


Fig 1: How machine/deep learning research is typically communicated *vs* an example of how the creative scientific process can go (red line). It is completely normal during the research process to rephrase the research question often; to revisit related work in a new light, to change the method or experimental setup, and re-analyze results and conclusions. Creatively searching for the question is inherent to science.

2 The storyline

I’ve developed the storyline-technique incrementally by reflecting on my experiences as a researcher. I find it one of the most valuable tools to anchor the creative process: to find, and work out, what question to concretely answer. The storyline is as detailed as possible, concise, focused, relevant, logical, self-contained, and fully-motivated research narrative, which can be understood and critiqued without the use of unnecessary jargon, and has all abstractions opened up to their concrete core reasons. It forms the heart of the question-search, and allows a holistic view on the full project, so that all aspects of the research can be scrutinized, questioned, critiqued, sharpened, removed, added, or pivoted on. It cuts everything away to arrive at the lean motivational core where any claim made can be challenged, and if it cannot be motivated, the claim should be removed. In empirical research, each claim can be challenged by an experimental question, and experiments thus take an important role and are tightly interwoven with each made claim. Since the storyline is for finding the question, it is therefore completely normal, and even expected, that the storyline will fluidly change during the creative process. Concretely, the storyline gives structure to the question search: thoughts, meetings, the process, the hypotheses, what experimental questions to ask, the logical narrative, and helps the writing by postponing sentence structure till later. A visualization of the storyline is shown in Fig 2.

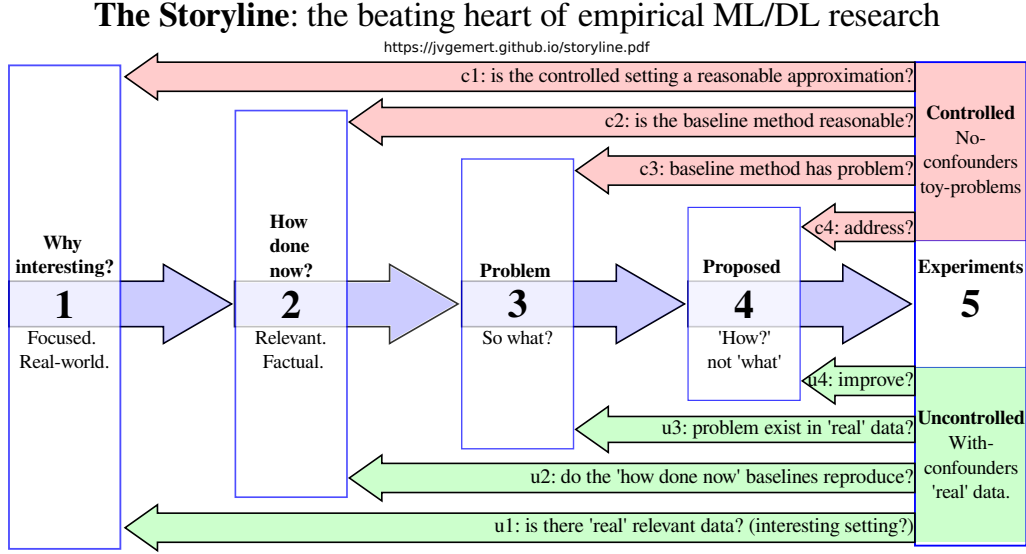


Fig 2: The storyline structures empirical machine/deep learning research. It concisely ties the research in a logically consistent focused narrative. It has the following elements. (1) Why the (real-world) setting is interesting; (2) factual description of relevant existing approaches (baselines); (3) what is the problem of the baselines and what are the consequences; (4) why the proposed approach addresses the problem and (5) experimental questions. The experiments take center stage in empirical research and they link back to all elements. One group of experiments are in a fully controlled 'toy problem' setting (top, in pink) with only a single confounder: precisely evaluate the settings with/without the identified problem. The controlled setting should empirically validate that (c1) the controlled experiment models reality sufficiently well, i.e.: it's unbiased; (c2) the baseline method (control setting) is set up reasonably and fairly, that this reasonable baseline (c3) does indeed suffer from the problem; and (c4) that the proposed approaches addresses it. Where the controlled experiments assume that the problem exists, the uncontrolled 'real' setting has unknown confounders (bottom, in green) and validate that the problem actually occurs in 'reality'. It shows relevance (u1) by demonstrating multiple relevant data sets; that (u2) published baselines reproduce published numbers; that the identified problem (u3) exists among uncontrolled confounders, and thus (u4) that the proposed approach improves over the published baseline when the problem exists.

2.1 Structural elements of the storyline.

This is the typical structure; each element might take 1-5 bullet points:

- (1). **Why interesting?** What is the tightly scoped motivation. Why should someone else care (e.g., the users of this particular research outcome).
- (2). **How done now?** Relevant approach(es) to the motivation in (1).
- (3). **What is missing, and So What?** What’s the problem with the approaches in (2), and what consequences does this have on (1).
- (4). **Proposed approach.** What do you do, and why does it address the problem in (3)?
- (5). **Experimental questions.** Controlled: validate that problem (3) occurs in current models (2) and that (4) addresses it, and its consequences (3). Uncontrolled: does the problem exists in confounding ‘real’ settings (1)?

The storyline is minimal, and stand alone: you cannot use a ‘jargon’ term/concept before it has been introduced by motivating it; i.e.: what is it and why is it needed. Each claim made can be challenged and each claim should thus be motivated. Keep Hitchens’s razor in mind: “*What can be asserted without evidence can also be dismissed without evidence*”. Terms/concepts logically build/connect to earlier terms/concepts (i.e.: its a story). Have short “1-liners” per bullet point; correct grammar is optional; the entire storyline should be visible at once, so it should fit on a single page/slide (10-20 bullet points). When finalizing the storyline, it’s useful to work backwards from the experiments; because the terms used there need to be introduced before. Remove unused terms and jargon.

Intermediate results produced during the research process often lead to a better understanding of the problem, and thus change the storyline, see Fig 1. After a while, there’s a collection of loosely connected results; and then its useful to add focus and re-evaluate which results fit a consistent narrative, and how this (new?) narrative changes the storyline and the follow-up experiments. Some experimental results will –in retrospect!– turn out to be distractors from the main narrative. Or, they were initial –now redundant– stepping stones towards a better understanding of the problem. These results play no role in the final paper: “*kill your darlings*”; which is common, but painful because often these results took quite some effort; and removing them seems to invalidate that effort, which, unfortunately, is inherent to the uncertainty of doing creative work.

2.1.1 Storyline element (1): Why interesting?

What is the motivation to do this research? Why should someone else care about this research’s outcome? Things that I often see are variants of “Much recent research is done on XXX”. These are not good reasons in themselves because it describes a reaction and not the reason for this reaction. Open the abstraction: i.e.: what are the underlying reasons that topic XXX has received so much attention? What is interesting, beneficial, useful, important, about it? Keep digging deeper and repeatedly ask “Why is that interesting?” until you cut away all abstractions and arrived at the concrete core. Put another way: why should someone invest time/effort in reading your work if it’s not clear what is interesting about it for them?

Keep the scope tight, and focus on your research outcomes. Your problem (3) and what you propose (4) should directly be applicable to the motivation in (1). For example, if the paper is about automatic reasoning in long videos, do not motivate it with ‘robots’, or ‘machine learning’, or ‘general visual recognition’, or even ‘action recognition in short clips’. Instead, try to motivate it directly and tightly focused with why automatic reasoning in videos is interesting; and what is specially important about ‘long videos’, which for example, could include sport game analysis, or shoplifting, and why doing automation is useful/interesting there. The scope and applications ideally come back in the experimental section. For example, a motivational scope claim on ‘robotics’ can (rightfully!) be asked for an experiment on an actual robot (evidence). Assessors can penalize unsubstantiated over-claiming, i.e.: tightly focus scope with problem/outcomes in (3,4).

As an example, consider a new method for visualizing biases in deep/machine learning models. If the claim is that the new method can find new types of biases then this claim can be challenged, and thus should be experimentally motivated by showing such new biases. As another example, consider a new optimization method that makes auto-encoders faster to train. Then if the claim is that auto-encoders are often used in applications such as denoising, feature-learning, super-resolution, etc., then this can be challenged, it thus experimentally shown that existing auto-encoder methods for such applications also actually become faster to train. Also for why the research is interesting, it holds that with each claim comes with a burden of proof.

2.1.2 Storyline element (2): How done now?

Here, give current, relevant, approach(es) to the tightly-scoped motivation in (1). Note the word *relevant* because the storyline is not meant to be exhaustive; instead, it's a focused, minimal and consistent narrative. Related approaches that are too different from the proposed approach, may belong in the 'related work' section of the research paper but not in the storyline, i.e.: they do not link to the problem (3) nor approach (4) and are thus not relevant for your story. For example, with a scope on 'long videos', the work on 'short videos' would go in related work, but is not relevant for the storyline. Leave out approaches that do not link to (3,4).

The described approaches here should be objective, without a value judgment. The authors of the approach(es) you mention should agree with how their approach is described; but the description does not have to be the main contributions of their work. You are free to creatively choose, re-interpret, and emphasize anything that is described in their papers, as long as it is factually correct. Be careful to not make use of jargon: each term used here should be motivated/introduced first. Moreover, terms can be introduced/motivated here so that they can be used in pointing out problems in (3).

2.1.3 Storyline element (3): What is missing, and So What?

What's the problem with the approaches in (2), and what consequences does this have. First describe the problem, or what is missing. Then, make the consequence of the problem precise. The consequences make it (experimentally) possible to validate that the problem occurs, that the baselines do indeed suffer from the problem, and that the proposed approach addresses the problem. For example, in a 'automatic long video recognition' setting, the way how it is done now (2) could be that models only sample one frame per minute (not true, but this is just an example). Then, a problem (3) could be that this low sampling rate might miss relevant information. And the consequences then are that current models are sensitive to the accidental sampling offset, and that they have low accuracy when higher-frequency information is essential. Thus, model rankings might not be correct, which would lead to selecting the wrong model in practice, which can be validated experimentally.

2.1.4 Storyline element (4): Proposed approach.

Why does your approach address the problem in (3)? Focus on the 'why' not on the 'what'. Avoid technical explanations as much as possible; the storyline is not about what the approach does, that does go in the 'method section' of the paper. Instead, the storyline is all about motivation, and building a logically consistent "house of whys". The proposed approach should be understandable for non-experts. Avoid jargon, if possible, because each specialized term used should first be motivated/introduced, either here, or in the preceding elements.

2.1.5 Storyline element (5): Experimental questions.

How do you evaluate experimentally that (4) solves the problem and its consequences in (3)? The focus here is on empirical machine/deep learning research, and experiments take center stage. See Fig 2 for how experiments build on each element. Not all research may require a storyline emphasizing the experiments as much; yet, the storyline itself is still valuable; feel free to flexibly adapt accordingly.

Typically the first line of experiments are for careful control and verification: validate if the problem with current approaches in (2) exists, how severe the consequences in (3) are, and how well the proposed approach in (4) deals with the problem and consequences. I advise self-made, fully controlled, synthetic, ('toy problem') setting, where the full control allows generating small, crisp, and precise variants, with known outcomes. This is important as to avoid unknown confounding variables which might, unknowingly, influence the results. One variant is a normal setting, without the problem i.e.: a control. This control setting demonstrates that the existing approaches are represented fairly (i.e.: they do OK), and to set a baseline performance. A second variant is identical to the first, where the only point it varies is that it has the identified problem in (3); which then demonstrates that existing approaches in (2) suffer from the consequences in (3) and that the approach in (4) is suitable. Note that "good accuracy" is not needed. I.e.: there is not need for large training sets, as that might actually be detrimental: if the baseline already scores 95%, then the proposed approach can only make marginal relative improvements.

A second line of experiments investigate impact in the world. Put another way: how severe is the identified problem, and its consequences in practice? The goal is to gather evidence that the problem occurs in 'reality', and it is good to have a couple of datasets as evidence. This involves evaluating on less controlled datasets, with unknown confounders, that have the problem. This can include real data that you collected yourself, or, existing open datasets. In academic research, such 'real world' datasets are typically still quite artificial. Even so, compared to the first line of experiments these datasets do have unknown confounders, and thus can be used as evidence that the problem exists. The datasets should align with the problem. e.g., if the problem involves rotated images, then typical Computer Vision datasets such as CIFAR, or ImageNet are out, because they do not contain rotations and are thus not relevant. Instead, use datasets where rotations occur naturally; for example cell images taken under a microscope. Here, it is important to validate that existing published results on the datasets actually reproduce on those datasets (do not assume they will!). The comparison to published results is important so that the reader can validate that baselines are fairly represented. These methods are the published baseline scores to compare to. If the problem occurs in the dataset, and the proposed approach handles the problem well, then it can be expected that the published baselines are outperformed by the proposed approach, on those datasets.

2.2 Collaboration through the storyline

The storyline forces a focused logical narrative that fits on a single page. Such a concise format facilitates communication because the whole narrative is visible and allows evaluating the full logical research structure by collaborators. If collaborators do not understand, or do not agree with the logic, then this is valuable feedback, and the storyline should be updated which corresponds to taking a step in search of the research question.

When discussing a storyline with its creator(s), I find myself typically in two different modes; in **construct mode**, we are together coming up with the storyline, often based on initial results, an initial method, or a research direction or problem. In **critique mode** I challenge the storyline as it is presented to me, typically by a detailed scrutiny of each word. This critiquing can be perceived as 'attacking' and might lead to unconstructive communication. I've found that if the storyline is still quite preliminary/incomplete/vague, then it's better to skip critique mode, and first go to construct mode; this

step can be initiated by both the feedback giver, but also the storyline creator. Once there is a more converged version of the storyline, then it becomes constructive to critique it's details. Note, it's important to be 'pedantic', because the logical narrative guides every part of the research. Also, it helps to discuss and go over the same storyline multiple times, because our future selves might see other things than our past selves.

Below are common problems that I encounter when discussing the storyline. Please use these as a self-check, after completing a storyline; they are also useful during meetings as anchor points.

Often re-occurring 'content' problems:

1. Not clear/precise why interesting or why people should care. Often the text is too general/abstract. Ideally, the reasons given come back in the experiments, and as such need to be concrete: we should be able to point at something that can concretely be done in an experiment.
2. Not clear/precise how it's done now: what are *relevant* current typical approach(es), often it is not specific enough.
3. Not clear/precise what is missing; what the problem is, and what consequences this problem has. Often the consequences are not linked back to the "why interesting". The consequences should be precisely the concrete things that the experiments will show that the proposed method addresses.
4. Not clear what you propose; what your method is/does.
5. Not clear/precise why your proposal/method could solve the problem. This is often either too general; or it is too technical. It should be at the "logical narrative" abstraction level.
6. Not clear/precise what experimental questions you ask that demonstrate solving a problem and addressing its consequences.
7. Includes irrelevant information: parts of the storyline can safely be removed without changing the main narrative.

Often re-occurring 'form' problems:

1. Using a term before defining/motivating it.
2. Too much unnecessary detail: words can be removed without significantly changing the storyline. Each word should have a reason to exist.
3. Unclear logical reasoning step.
4. Inconsistent use of terminology. Use a single term for a single concept; 1-to-1, consistently.

5. Writing is too verbose and too many full sentences are used: bullet point should be short one-two lines; correct grammar is optional.
6. Too many topics per storyline bullet point. Keep a single point on a single topic.
7. Too many storyline bullet points. Keep the storyline focused. Less is more. Find the minimal logical story for the research project.
8. The text is not stand-alone; it's not peer understandable. Too much jargon, or 'suitcase words'.
9. Not following my writing guidelines: <https://jvgemert.github.io/writing.pdf>;

2.3 Storyline examples

To make the storyline structure as described above less abstract, I give some example storylines for a few papers where I was involved, below. Note, they are not meant to be perfect; but the real world rarely is perfect and “*perfection is the enemy of good enough*”. The storyline is a tool to be used flexibly. In reality there often are time constraints (work contracts, graduation dates, etc.). Science is never 'done', and a scientific paper can still be interesting (ie: publishable) when it is 'good enough'. In addition, some research project have different empirical questions that do not fully align with the controlled/uncontrolled experimental groups in (5); which is fine. The power of the storyline is the harsh logical narrative, that forces the researcher to back up a claim with evidence; or adapt the claim accordingly.

Storyline for Lengyel, et al. *Color Equivariant Convolutional Networks*, NeurIPS, 2023. <https://arxiv.org/abs/2310.19368> (annotated storyline illustrating the logical narrative built on terms that are first motivated before they are used.)

1. **Why interesting?**

- (a) Automatic image recognition is important for many applications.
- (b) Image recognition is trained on data with inherently imbalanced (accidental) viewpoint/appearance occurrences.
- (c) Imbalance leads to biases towards the frequent; and reduced accuracy for the less frequent occurrences.

2. **How done now?**

- (a) Imbalance is tackled by Equivariant CNNs: sharing learnable weights over spatial transformations (rotations, scale, ..).

3. **What is missing, and So What?**

- (a) Current equivariance work is on spatial transformations, no appearance.
- (b) So: reduced accuracy due to imbalance in appearance.

4. **Proposed approach.**

- (a) Focus on color (hue) as a case of non-spatial appearance.
- (b) Sharing weights over different appearances: color hues (hue = H in HSV color space).
- (c) Propose: color equivariance by rotations in hue space.

5. **Experimental questions.**

- (a) Gains for class/color imbalance? Toy set: *Long-tailed ColorMNIST* has 30 classes (10 digits x 3 colors), controlled imbalance.
- (b) Gains for color transformations? Toy set: *Biased ColorMNIST* has 10 classes (digits), give each sample a random color; create a curve over color transformations by varying the stddev of the random color.
- (c) Gains for existing, color oriented, datasets?

Storyline for Kayhan et al. *On Translation Invariance in CNNs: Convolutional Layers Can Exploit Absolute Spatial Location*, CVPR, 2020. <https://arxiv.org/abs/2003.07064>

1. Why interesting?

- (a) Automatic visual classification, image matching, video recognition are important for many applications.
- (b) Sharing network weights over different locations (spatial shift equivariance) improves data-efficiency:
- (c) data-efficiency is important: collecting/labeling data is expensive.

2. How done now?

- (a) CNNs use convolution (sliding window) to share weights over different locations.
- (b) When the sliding window reaches the image boundary it stops, or, half the window-size zeros are padded outside the image boundary.

3. What is missing, and So What?

- (a) Insight: input pixels near the image boundary are not seen by the full sliding window (it: it slides only until the image ends).
- (b) Ie: CNNs can asymmetrically ignore image content close to one side of the image boundary and not the other side.
- (c) Surprisingly: CNNs can learn weights that depend on the location, by the distance to image boundary.
- (d) So: when shift-equivariance is broken; data efficiency suffers; needing more expensive labelings.

4. Proposed approach.

- (a) Remove the ability of the model to exploit absolute location:
- (b) Make the sliding window see all input pixels (ie, the left part of the window should also see all pixels of the right part of the image)

5. Experimental questions.

- (a) Can 1 CNN layer use location? Train CNN to separate 2 images; each with the identical patch, but at different spatial locations.
- (b) How far from the image boundary can existing (scratch/pre-trained/random) CNN models use location? Same experiment as in (a) but vary the distance of the patch to the image boundary
- (c) Will baseline/proposed use location when not needed? For location-independent task: train on one location; test on a different one.
- (d) Sensitivity to spatial shifts. Evaluate on test-time image shifts.
- (e) Data-efficiency? Learning curves: classification, matching, video.

1. Why interesting?

- (a) To train a ML model we need to label/annotate data: boring, expensive, time consuming, and error prone (annotation noise).
- (b) 'Active Learning' reduces the label effort by not labeling the full dataset: Use ML model trained on partial labels during the labeling: interactively suggest 'most informative' data samples to label by a human annotator; retrain; repeat.

2. How done now?

- (a) Various active learning strategies to select the samples to label

3. What is missing, and So What?

- (a) Labeling samples focuses on the data points; but the goal is to find the decision boundary between classes.
- (b) So: labeling samples not directly solving the goal.
- (c) So: labeling samples will take more label effort than if we could instead label the decision boundary directly.

4. Proposed approach.

- (a) Instead of labeling samples; lets label the decision boundary itself.
- (b) Use a generative model based on baseline active learning sample strategies to generate a 'line' of samples which crosses the decision boundary.
- (c) Let the user annotate on the 'line' where a class changes to a different class: this is where the decision boundary lies.
- (d) The ML model can then include decision boundary annotations.

5. Experimental questions.

- (a) How well can baseline active learning sample strategies be used as input for decision boundary annotation?
- (b) Quality of generative model close to the decision boundary?
- (c) Sensitivity to noisy decision boundary labeling?
- (d) How well does a human annotator do with decision boundary annotation?
- (e) How well does it generalize to more classes/datasets?

Storyline for De Boer et al. *Is there progress in activity progress prediction?*, ICCVw, 2023. <https://arxiv.org/abs/2308.05533>

1. Why interesting?

- (a) Action progress predictions useful for scheduling, planning.

2. How done now?

- (a) Current methods aim to learn visual information to predict action progress.

3. What is missing, and So What?

- (a) Published visual-learning methods never compare to simple baselines.
- (b) So: Unclear if visual-learning methods "work".
- (c) So: Unclear if visual-learning methods can be trusted, or should be used in reality.

4. Proposed approach.

- (a) Set 2 simple visual learning baselines: 'CNN', and a 'CNN+LSTM'.
- (b) Set 2 simple non-visual learning baselines: 'frame-counting' and 'random-noise as input'.
- (c) Set 2 non-learning baselines: 'random guessing'; and 'always predict 0.5'.
- (d) Create a synthetic dataset: 'visual progress bar' to evaluate if current visual-learning methods can do progress prediction.

5. Experimental questions.

- (a) Evaluate 3 existing datasets with: 3 published visual learning models; 2 simple visual-learning baseline models; 2 non-visual baselines (frame-counting, random-noise input) and 2 non-learning baselines (random guessing, always 0.5).
- (b) Evaluate taking segments instead of full videos, to try to avoid frame counting, because learning from a segment with it's progress score does not have the full-video context.
- (c) Evaluate progress prediction methods on visual 'progress bar'.

Storyline for Strafforello et al. *Are current long-term video understanding datasets long-term?*, ICCVw, 2023. <https://arxiv.org/abs/2308.11244>

1. Why interesting?

- (a) Long-term automatic video understanding: sports, surveillance.

2. How done now?

- (a) Quality of methods are evaluated on long-term video datasets.

3. What is missing, and So What?

- (a) Unclear if current long-term video datasets really evaluate on long-term information.
- (b) So: methods that do well on these datasets might not do well on actual long-term settings.
- (c) So: might lead to bad results in reality; wasted costs; disappointed users; failed projects.

4. Proposed approach.

- (a) Define: Long-term must consist of multiple short-term actions.
- (b) Evaluate if humans can recognize long-term actions in video datasets after seeing a short clip. If so, then the videos are not long-term.

5. Experimental questions.

- (a) For datasets that have both long-term and short-term annotations, there should be more than 1 short-term actions annotation used in a long-term action annotation. If not, then long-term can be recognized by a short-term; and it's therefore not long-term.
- (b) For several long-term datasets: create 2 sets for the same videos. A set short-segments and a set of long-segments; validate for a long-term task that $\text{accuracy}(\text{long-segment}) > \text{accuracy}(\text{short-segment})$; if this is not true, the videos are not 'long-term'.

References

- [1] Rick Rubin. *The Creative Act: A Way of Being*. Penguin Random House, 2023.
- [2] Itai Yanai and Martin J. Lercher. What is the question? *Genome Biology*, 20, 2019.